

SoDam O-Brain — AI Memory System

A 100% local memory tool that automatically saves important decisions and notes from your AI conversations, and visualizes them as a knowledge graph.

In one line: AI is brilliant but forgets everything the moment a chat ends. O-Brain is the **auto-notepad** you keep next to that AI — it writes down what matters and brings it back whenever you need it.

Key Features

Feature	Description
100% Local	All memories stay on your PC only. No external server or cloud transfer
Hybrid Search	Keyword (FTS5) + semantic similarity (vector search) simultaneously
Knowledge Graph	Visualize memories as 2D/3D dots and lines
Time Travel	Query memories that were "alive" on any past date
Confidence Decay	AI-extracted memories auto-decay with a 30-day half-life
Security Filter	API keys and passwords auto-removed before saving ([REDACTED])
Auto Backup	Snapshot created automatically on server start and before bulk delete
MCP Integration	7 tools for saving/searching/linking memories directly from Claude Code
Orphan Node Visual	Memories with no user-defined relations shown with a dashed ring

Prerequisites

Item	Requirement
OS	Windows 10 / 11 (64-bit)
Node.js	20.x LTS or higher (free from nodejs.org)
Storage	500 MB+ (includes ~90 MB embedding model)
RAM	4 GB+ (8 GB+ recommended)
Browser	Chrome, Edge, Firefox, etc.

Claude Code is only required for MCP integration. The web dashboard works without it.

Installation (2 Steps)

Step 1: Install dependencies (in a terminal)

```
cd D:\WAI_Dev_Work\2026y\26y_06m_21d_SoDam_O-Brain\Wapp
npm install
```

Step 2: Start the server

```
npm start
```

→ When you see 0-Brain Local Server ► http://127.0.0.1:7740 , it's running.

Open in browser:

```
http://127.0.0.1:7740
```

First run: automatically downloads the AI embedding model (~90 MB). Takes 1–3 minutes. All subsequent runs start instantly.

Claude Code Plugin (Optional)

If you use Claude Code, you can also install it as a plugin:

```
/plugin marketplace add D:\AI_Dev_Work\2026y\26y_06m_21d_SoDam_0-Brain\plugin
/plugin install o-brain@o-brain-local
```

Restart Claude Code after installation → slash commands become available:

Command	What it does
/o-brain:status	View current memory count and status
/o-brain:selftest	Run save/search/security checks (✅✅✅ = OK)
/o-brain:remember	Extract and save decisions from current chat
/o-brain:link	Link memories with a relation
/o-brain:backup	Create a manual backup
/o-brain:open	Open the dashboard in your browser

MCP Tool Integration (Claude Code)

Register in Claude Code's settings.json to save/search memories mid-conversation:

```
{
  "mcpServers": {
    "o-brain": {
      "command": "node",
      "args": [ "C:/absolute/path/to/app/src/mcp-server.mjs" ]
    }
  }
}
```

Available MCP tools: save_memory , search_memory , get_memory , get_related , get_timeline , add_relation , list_categories

Key Commands

```
# Run from the app/ folder
npm start           # Start server (port 7740)
npm run selftest    # Run self-diagnostics
npm run backup      # Manual backup
npm run status      # View DB stats
npm run seed        # Add sample data (for testing)
```

File & Data Locations

Item	Location
Memory database	app/data/obrain.db
Auto backups	app/data/backups/ (keeps latest 10)
API token file	app/data/.api-token (regenerated each run)
Personal settings	app/.env.local (uses defaults if absent)

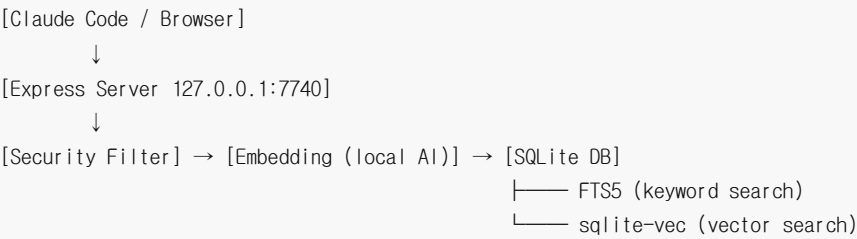
Environment Variables (app/.env.local)

Variable	Default	Description
OBRAIN_PORT	7740	Server port number
OBRAIN_DATA_DIR	./data	Data storage folder

Security Overview

- Server binds **only to 127.0.0.1 (your PC)** — not reachable from external devices
- **API keys and passwords auto-removed** before saving (processed by redact.mjs)
- New **local API token** auto-generated each server start (crypto.randomBytes)
- `data/` , `.env.local` , `*.sqlite` are in `.gitignore` — never committed to Git
- No external cloud communication. Embedding model runs entirely locally

Architecture Summary



Troubleshooting Quick Reference

Symptom	Fix
'node' is not recognized	Install LTS from nodejs.org → open a new terminal
EADDRINUSE :::7740	Add OBRAIN_PORT=7741 to .env.local and restart
Graph is empty	Run npm run seed to add sample data
Browser won't connect	Confirm npm start is running → check address is http://127.0.0.1:7740
Embedding download fails	Check internet connection · allow Node.js through firewall
Accidentally deleted a memory	Restore from app/data/backups/ latest .db file

Full guide: [GUIDE.en.md](#) (English) · [GUIDE.md](#) (Korean)

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